Homework 9

Problem 1

Consider the Bernoulli regression model,

$$P(y_i = 1) = p_i = \frac{e^{x_i\beta}}{1 + e^{x_i\beta}}, \quad y_i \in \{0, 1\}, \quad i = 1, \dots, n,$$

with β a one dimensional unknown parameter.

(1) (10 pts) Write down the log likelihood function $L(\beta)$ for β.

(2) (10 pts) Find both $dL/d\beta$ and $d^2L/d\beta^2$.

(3) (10 pts) Write down a Newton–Raphson algorithm for finding the maximum likelihood estimator $\widehat{\beta}$.

(4) (10 pts) If $\widehat{p_i} = \exp(x_i\widehat{\beta})/(1+\exp(x_i\widehat{\beta}))$, show how to obtain the approximation $E(\widehat{\beta}) = \beta$ and

$$\mathrm{Var}(\widehat{\beta}) = \frac{1}{\sum_{i=1}^n x_i^2 \,\widehat{p_i}(1 - \widehat{p_i})}.$$

(5) (10 pts) Hence, describe a test for the hypothesis $H : \beta = 0$.

Problem 2

The Bernoulli regression model, as given in Problem 1, is analyzed using a Bayesian approach and the prior for β, i.e. $\pi(\beta)$, is chosen to be normal with mean 0 and variance σ^2.

(1) (10 pts) Write down the posterior density (up to a constant of proportionality) for β in terms of the (x_i, y_i).

(2) (10 pts) A way to sample from a density directly is available if the logarithm of the density is concave. Show that the log of the posterior density from part (1) is concave.

(3) (10 pts) If the prior is taken to be improper, i.e. $\pi(\beta) \propto 1$, which would follow if σ is taken to infinity, what is the relationship between the posterior and the maximum likelihood estimator.

(4) (10 pts) With this improper prior, the aim is to derive the Bayes factor for the choice of null hypothesis $H_0 : \beta = 0$ versus the alternative $\beta = 1$. Show that the log Bayes factor can be written as

$$\log B = \sum_{i=1}^n x_i y_i - \sum_{i=1}^n \log\left(\frac{1 + e^{x_i}}{2}\right).$$

(5) (10 pts) Find the approximate normal distribution of $\log B$ under the null hypothesis.